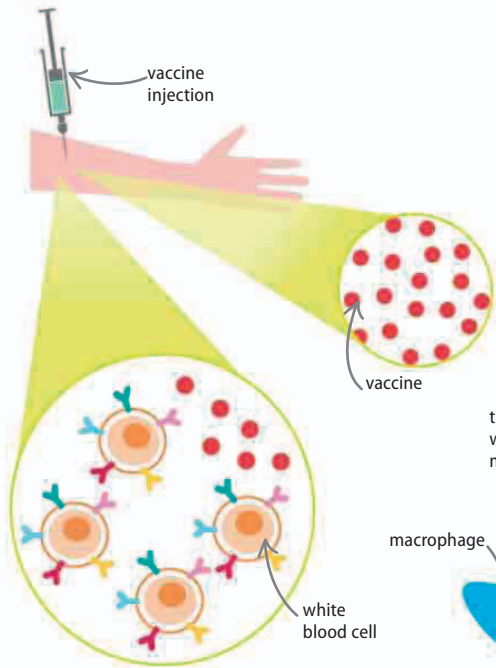
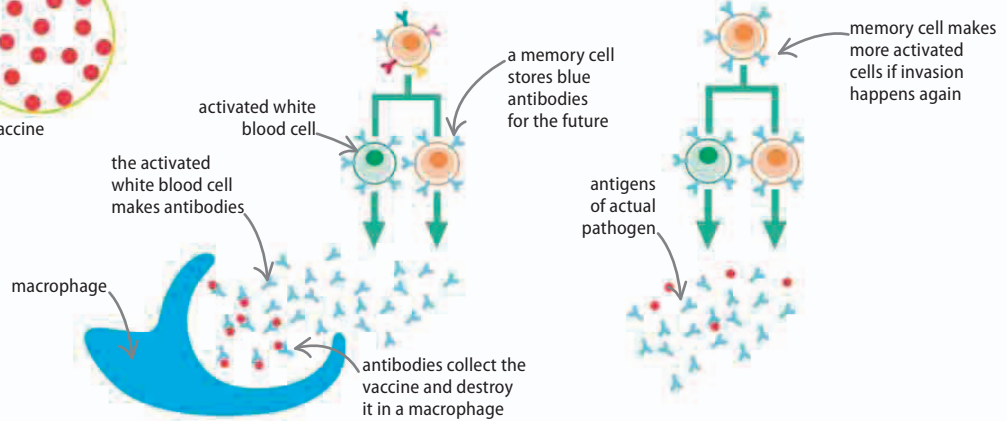




Vaccination

The immune system is exploited by doctors to protect people from disease using vaccines. A vaccine introduces the antigens of a pathogen to the body—either in a chemical form, or a weakened strain that does not cause debilitating symptoms. The body responds to these antigens, and, if the actual pathogen enters the body at a later date, the immune system recognizes it and immediately kills it.



△ 1. Vaccination

The vaccine is injected into the body, where it is detected by the white blood cells just like any other foreign antigen entering the body.

△ 2. Antibodies

The white blood cells develop antibodies for the vaccine, which kill the vaccine antigens. The antibody is then stored in a memory cell.

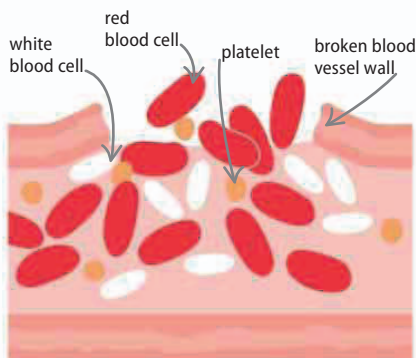
△ 3. Fighting infection

The real pathogen has the same antigen as the vaccine. If it enters the body, the immune system deploys the stored antibodies and the disease does not develop.

Healing skin

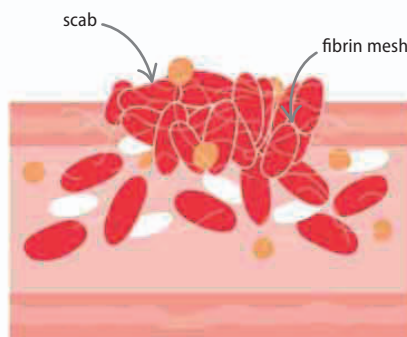
The body's first line of defense against attack is the skin. When the skin is broken by a cut or scrape, bacteria and other germs can get into the body. Therefore, blood rushes to the area, making it swell and helping seal the gap. The liquid blood quickly coagulates (thickens) into a solid scab that forms a temporary seal while the skin grows back.

Sufferers of **hemophilia** have a reduced ability to form blood clots, which can result in a small scrape causing them to bleed to death.



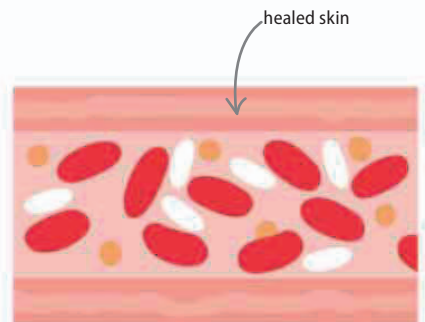
△ 1. Broken skin

The blood floods into the gap in the skin. Tiny cells called platelets react to the skin proteins. They trigger swelling, which brings white blood cells to mop up any invaders.



△ 2. Coagulation

The platelets release the enzyme thrombin, which converts a soluble protein called fibrinogen into an insoluble one, fibrin. The fibrin forms a solid mesh across the gap.



△ 3. Healing

The temporary seal, or scab, lasts until the skin has grown back. When this happens, the inflammation reduces, and the fibrin dissolves back into the blood.